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opheliaayamga246810-arch Update ophelia-week6-day2.md dfd83be · now

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Ophelia: Week 6, Day 2

5 Step Strategy To Selecting, Training and Applying an LLM to a Commercial Problem

5 Step Strategy

1. Understand
2. Prepare
3. Select
4. Customize
5. Productionize

1. Understanding

Activities -

1. Gather business requirements for the task
2. Identify performance criteria
    - Model and Business centric metrics
3. Understand the data: quantity, quality, format
4. Determine non-functionals:
    - Cost Constraints, Scalability, Latency
    - R & D / build budget and implementation timeline.

2. Preparation

Activities

a. Research existing / non-LLM solution

- Potential baseline Model. b. Compare relevant LLMs.
- The basics, including context length, price and [Tokens]
- Benchmarks, Leaderboards, and Arenas &
- Specialist scores for the task at hand c. Curate data: clean, preprocess and split

3. Selection

Activities

- Choose LLMs
- Experiment
- Train and validate with curated data.

4. Customization

Technique to optimize the performance of the models

1. Prompt
2. RAG
3. Agents
4. Fine-tune

- Prompt -> using prompts to bias the model to making model's output consistent with input
- RAG - Looking for ways to select relevant context, e.g. using vector Market similarity, Embedded model.
- Agents (Agentic AI) - Breaking Extension of prompt, doing LLMs calls into Multi calls and having one of the calls decide which to call and in what order
- Fine-tune - giving LLMs data to make it better at a task
- The first 3 are all Inference Time Techniques - levers, run time you used to train Models to get most out of them
- The last one is a Training Time Technique - The actual training of the Model
- Mostly, more time is spent on the Inference Time Section.
- RAG - Ideal for Generalisation. Generalise Models
- Fine tuning - Ideal for Models built purposely for one area of expertise

5. Productionize

Activities

1. Determine API between model and platform(s).

2. Identify model hosting and deployment architecture.
3. Address scaling, monitoring, Security, compliance, observability.
4. Eval: measuring the Business - focused Metrics identified in Step 1
5. Continuously retrain and measure performance.